CHANAKYA Project Progress Report

For Consideration Under

TIH – IoT and IoE

Development of an Image-based disease detection leveraging AI/ML techniques and Computer Vision

Submitted by

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Revision History

Rev.	Name	Issue Date	Changes
1.0	Project Milestone Evaluation Report	30th April 2026	1st Milestone Evaluation Report

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1 Cover Sheet

Sr. No.	Name	Details
1	Principal Investigator (PI)/ Fellows name	Aakash Trimbak Jawle Kamaaksha Rajendra Kharul
2	CHANAKYA Fellowship Project No	TIHI-149468
3	CHANAKYA Grant Number	TIH-IoT/2025-11/HRD/CHANAKYA/SL/CFP/TIHI-149468
4	Project Title	Development of an image-based disease detection leveraging AI/ML techniques and computer vision
5	Date of Project Commencement	01 December 2025
6	Date of Project Review	30th April 2026
7	Progress Report Serial No	1
8	Reporting Period	December 2025 – April 2026

2 Introduction

Agricultural productivity is significantly affected by plant diseases, leading to reduced crop yield and economic losses. Early and accurate detection is essential for timely intervention. Traditional approaches rely on manual inspection or image classification models, which only provide disease labels without interpretability.

To address this limitation, this project focuses on developing an advanced **Vision-Language Model (VLM)** system capable of not only identifying plant diseases but also generating **structured and explainable captions**. The model

outputs include plant type, disease condition, severity level, pathogen type, and visible symptoms, making the system more informative and practically useful.

The work carried out during the reporting period followed a structured progression:

- **Phase 1 (December–February):** Fundamental image processing, CNN-based models, and transfer learning
- **Phase 2 (April–May):** Vision-Language Models, multimodal learning, structured caption generation, and deployment

The project transitioned from conventional computer vision techniques to advanced multimodal AI systems, enabling explainable disease detection.

3 Milestone's Review

Particular	Details
Aims	1. Review presentation and progress report to ensure sufficient progress has been made. 2. Review effort, technical achievements etc.
Requirements	Oral Presentation: 20-min presentation + 10min questions.
Documents to be submitted	Project report consisting of Introduction; hypotheses/aims; progress report (results, discussion, conclusions); plans for future work (including a timeline); materials/methods; references; list of publications and presentations (papers and abstracts); <30 pages. The report should be submitted one week prior to the date of review.
Outcomes	Guidance on project direction, identification of areas for Improvement and prioritization.

Review frequency	The milestones will be reviewed after every six months.

3.1 List of Milestones as per the Approved Project

Sr. No.	Milestone	Target Date of Completion	Actual Date of Completion	Status
1	Image Processing & Preprocessing Pipeline Development	Dec 2025	Dec 2025	Completed
2	CNN Training & Transfer Learning Models	Jan 2026	Jan 2026	Completed
3	ML Pipeline Development & Deployment Testing	Feb 2026	Feb 2026	Completed
4	Vision-Language Model Fine-Tuning & Evaluation	Apr 2026	Apr 2026	Completed

3.2 Shortfalls and reasons in Achievements of Milestones for the Review Period

- Flash Attention optimization could not be implemented due to incompatibility between CUDA version and PyTorch environment
- Initial instability and architectural mismatches while working with LLaVA-based models
- Dataset inconsistencies due to varied folder naming conventions requiring additional preprocessing logic
- High computational requirements of Vision-Language Models leading to longer experimentation cycles
- Token length limitations caused truncation issues in structured caption generation

3.3 A brief Description of Technical/Scientific Achievements for the Review Period with design/image of prototype)

- Designed and implemented a complete end-to-end AI pipeline, starting from raw image acquisition to final deployment, including data ingestion, preprocessing, model training, evaluation, and real-time inference modules
- Implemented classical image processing techniques using both OpenCV and custom NumPy-based convolution, including Gaussian smoothing, Sobel edge detection, Canny edge detection, and interpolation (nearest neighbor and bilinear), enabling low-level understanding of feature extraction
- Developed a modular preprocessing pipeline incorporating resizing, normalization (mean-std scaling), color space standardization, and cropping, ensuring consistent input distributions across training and inference
- Integrated data augmentation strategies such as brightness variation, Gaussian noise injection, and blur simulation to improve generalization and robustness against real-world variations
- Trained and benchmarked multiple CNN architectures including ResNet34/50, EfficientNetB3, MobileNetV2, and ConvNeXt Tiny, analyzing trade-offs between accuracy, computational cost, and inference latency
- Implemented transfer learning using pretrained ImageNet weights, fine-tuning deeper layers while freezing initial layers to optimize performance on limited agricultural datasets
- Built a scalable training and validation pipeline with proper dataset splitting, batch loading using PyTorch DataLoader, and performance tracking across epochs
- Developed and validated a crop-specific soybean disease detection model, focusing on improving robustness and performance on soybean datasets through targeted preprocessing, training, and evaluation
- Extended the system to support multi-crop evaluation, testing across datasets including soybean, cotton, and sugarcane to ensure cross-domain generalization and robustness

- Transitioned from classification models to multimodal Vision-Language Models (VLMs) to address the limitation of non-explainable predictions
- Engineered a custom structured caption dataset from PlantVillage images by parsing folder hierarchies and generating semantic labels including plant type, disease condition, severity, and symptom descriptions
- Developed heuristic-based mappings for disease severity and visual symptom patterns, enabling consistent caption generation aligned with domain knowledge
- Fine-tuned multiple VLM architectures including BLIP-2 (OPT-based), LLaVA-1.5/Next (Mistral backbone), and Qwen2.5-VL, evaluating their performance in multimodal reasoning and structured output generation
- Applied Parameter-Efficient Fine-Tuning (PEFT) using LoRA, targeting attention projection layers (q_proj, v_proj, k_proj, o_proj) and multimodal projection modules, significantly reducing trainable parameters
- Utilized bfloat16 mixed precision training and gradient accumulation to efficiently utilize approximately 22–24GB VRAM on NVIDIA RTX 4500 Ada GPU
- Attempted Flash Attention optimization for faster training but encountered CUDA compatibility limitations, leading to fallback optimization strategies
- Designed structured prompts for instruction tuning, enforcing output format consistency (Plant, Condition, Severity, Description) within VLM outputs
- Implemented a comprehensive evaluation framework, including BLEU-1, BLEU-4, ROUGE-L, CLIP similarity, accuracy, precision, recall, F1-score, R^2 score, and Mean Absolute Error (MAE) for severity estimation
- Developed a Streamlit-based deployment interface enabling real-time image upload, caption generation, and structured disease explanation
- Conducted extensive application-level testing across multiple crops and multiple application versions to ensure stability and usability
- Created a complete system demonstration video showcasing end-to-end workflow, real-time inference, and multi-crop performance validation

3.4 Patents filed during the Review Period

N/A

3.5 Paper publication/submission in Journal/Symposia/Conferences during the Review Period

Title of the paper	Journal/Symposium/Conference	Date of conference	Authors
NIL	NIL	NIL	NIL

3.6 Any other Comment of Significance

- Developed a crop-specific soybean disease detection model with targeted evaluation and validation
- Tested a real-time application interface, validating usability across different datasets
- Created a complete demonstration video illustrating system workflow, inference process, and multi-crop capabilities

3.7 Any technical guidance/support needed from TIH office

- Guidance on converting project into research publication

3.8 Action taken on the Observations of the Project Review Committee in its preceding Meeting

Sr. No.	Observation	Action Taken
1	Improve model performance	Implemented transfer learning and VLM fine-tuning
2	Increase dataset quality	Developed structured caption dataset with augmentation

3.9 Milestones to be achieved for next six to 12 months

- Fine-tuning larger VLM models (13B and above)
- Expansion to additional crop datasets
- Research paper publication and patent exploration